LESSON PLAN

AVIONICS AND ELECTRONIC WARFARE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	16
Total No. of Lesson Plans	25	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Purpose & operation of A/C Internal lighting: - Instrument illumination lamp, control panel light, cabin lights, map lights, emergency escape light passenger information light
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Purpose & operation of A/C Internal lighting: - Instrument illumination lamp, control panel light, cabin lights, map lights, emergency escape light passenger information light. Practical on position & wiring of lights.
5. Specific Objectives	13.1.3.1 Knowledge on circuit protection devices, generators, actuators & engine starting system.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Practical on position & wiring of lights; Safety: Verify power off before handling; Supervision: Instructor monitors wiring connections and circuit protection verification.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of internal aircraft lighting?
(b) Name three types of internal lighting systems.
(c) Where is the main lighting control panel typically located?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Instrument illumination lamp operation • Control panel light functionality • Cabin lighting system overview • Map light positioning and use • Emergency escape light activation	L & Ds	PPT, IB	15 min
(b)	• Passenger information light operation • Circuit protection devices for lighting • Generator supply to lighting systems • Actuator mechanisms for light control • Engine starting system impact on lighting	L & Ds	PPT, PDF	20 min
(c)	• Practical identification of light positions • Wiring inspection and verification • Circuit breaker location and testing • Emergency light battery check • Control switch operation drill	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between normal cabin lighting and emergency escape lighting.
(b) Describe the role of circuit protection devices in the internal lighting circuit.
(c) How does the engine starting system affect the availability of internal lighting?

12. Home Assignment and Follow Up	Prepare a brief note on the wiring layout and circuit protection for the next lesson on external lighting systems.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Purpose & operation of A/C external lighting: - Navigation lights, anti-collision lights, landing lights and taxing lights, Ice inspection lights practical on position and wiring of lights.	Refer to the master syllabus reference list	LP 17

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min